Next-Day Direction of GLD

Lagged Prices, Feature Selection, and a Hold-Out Test

MSDS 451 Financial Engineering
Programming Assignment 1
Leandro 11 October 2026

Abstract

I test whether lagged daily prices forecast the sign of the next close-to-close return on GLD, the SPDR Gold Shares ETF. The design follows the course jump-start: fifteen price features, an all-subsets search by the Akaike information criterion, and an XGBoost classifier tuned inside a time-series cross-validation. I add the step the jump-start leaves out. Features and hyperparameters are chosen on a development window of 4,515 sessions, from 2006-01-09 through 2023-12-14. A ten-session gap is discarded. The fitted models are scored once on 696 later sessions, from 2024-01-02 through 2026-10-09. A second feature set replaces price levels with returns and ratios, because a drifting price is a poor input to a classifier. On the hold-out, the price-level model is right on 43.2 percent of days (AUC 0.532). The stationary model is right on 55.2 percent of days (AUC 0.501). Calling every hold-out day an up day is right on 56.9 percent of days. Lagged prices do not identify the next day's direction in this sample.

Introduction

The jump-start program predicts whether West Texas Intermediate closes up or down, then reports accuracy on the same rows used to choose the model. The course note on model development asks for a different test: fit on one stretch of time, and score a later stretch that played no role in the fitting. I keep the jump-start's tools and change the asset and the test.

The asset is GLD. The assignment bars WTI and asks for a series other than the example. Gold is a monetary metal, not a barrel of oil, and the ETF has a long daily history on Yahoo Finance. The question is the one the example poses. Do a handful of lagged price features tell us the sign of tomorrow's return?

The week's stationarity lecture is the reason for a second model. A price that trends is not a usable signal. The log return is. I estimate both feature families under one selection rule and one classifier, and I compare both with two rules that ignore the features.

Literature

Akaike (1973) supplies the selection score: a penalty for each extra parameter, and a reward for a higher likelihood. Hastie, Tibshirani, and Friedman (2009) treat an enumeration of subsets as more trustworthy than stepwise search when the subset space is small. Fifteen features produce 32,767 non-empty subsets, which a logistic regression can score in one pass. Geron (2022) is the reason the XGBoost settings are drawn at random rather than laid out on a grid. Lopez de Prado (2018) is the reason the published score comes from a later window, with a gap so that a moving average does not sit on both sides of the cut. Chan (n.d.) states the stationarity point in the module lecture: machine learning wants stationary inputs, and the time difference of the log price is the usual repair. Ross (2011, chaps. 5 and 8) is the arbitrage reading for the week. A rule that called the sign of tomorrow's gold return often enough to survive trading costs would be a persistent edge of the kind those chapters say the market removes. This paper stops one step earlier. It asks whether the sign is

forecastable at all.

Methods

Data and target

Daily bars come from Yahoo Finance through the yfinance package, ticker GLD, split-adjusted, from 3 January 2006 through 9 October 2026 (5,225 sessions). Building four lags drops four rows and leaves 5,221. The target is 1 when the close-to-close log return is positive and 0 otherwise. Every feature is known at the previous close. The same day's open, high, low, close, and volume are excluded, so the model cannot see the move it is asked to call.

The development window runs from 2006-01-09 through 2023-12-14 (4,515 sessions). The next 10 sessions, 2023-12-15 through 2023-12-29, are dropped. The hold-out runs from 2024-01-02 through 2026-10-09 (696 sessions). Up days are 52.4 percent of the development window (2,368 up, 2,147 down) and 56.9 percent of the hold-out (396 up, 300 down). The hold-out is the gold rally of 2024-2026. I do not rebalance the classes. The ROC summary does not require equal class counts.

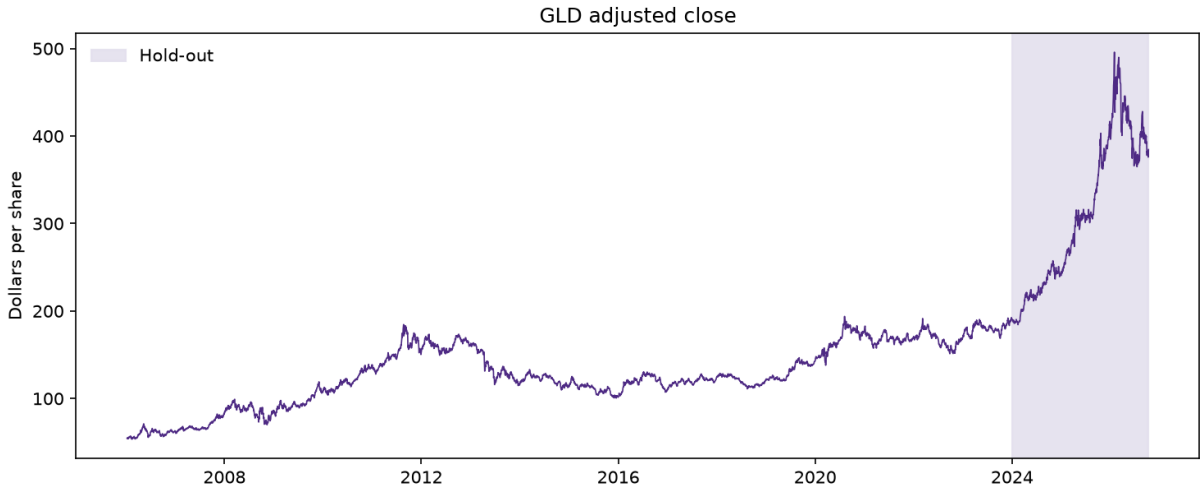


Figure 1. Adjusted close of GLD. The shaded band is the hold-out. The development maximum close is $193.89. The hold-out median is $305.46, and 654 of 696 hold-out closes sit above every development close.

Two feature families

The price-level family is the jump-start list. Close lags of one, two, and three days. High minus low, and open minus close, at the same three lags. Volume at the same three lags. Exponential moving averages of yesterday's close with half-lives of one, two, and four days, labeled CloseEMA2, CloseEMA4, and CloseEMA8 to match the example. The averages are taken on the lagged close, so today's close does not enter them.

The stationary family replaces levels with changes. Three lags of the log return. High-minus-low and open-minus-close, each divided by the close on that same day, at three lags. Three lags of the log change in volume. Exponential moving averages of yesterday's log return, with the same three half-lives. An Augmented Dickey-Fuller test on the development window fails to reject a unit root in the close (statistic -1.56, p = 0.50) and rejects one in the log return (statistic -68.0, p reported as zero at machine precision).

Selection, tuning, and the hold-out rule

On the development window only, each family is standardized and every non-empty subset is fit with a logistic regression. The Akaike score is 2k minus twice the in-sample log likelihood, and k counts the slopes plus an intercept. A feature is kept when it appears in at least five of the ten lowest-scoring subsets. That rule is fixed before looking at the hold-out. It kept four price features and two stationary features.

The classifier is XGBoost with a logistic objective and histogram trees. Inside the development window, a five-fold TimeSeriesSplit with a ten-day gap scores a first model of 200 trees. Randomized search then draws 30 parameter vectors from max depth 2 to 7, minimum child weight 1 to 9, subsample 0.5 to 1, learning rate 0.01 to 0.11, and 80 to 300 trees. The draw is seeded at 2026. The search maximizes ROC AUC. The jump-start maximized accuracy and then drew its ROC from hard class labels. AUC uses the probabilities, which is what a later cutoff would threshold. The scaler sits inside the pipeline, so each fold scales on its own training rows. The winning settings are refit on the whole development window.

The hold-out is scored once, with an up call when the predicted probability is at least one half. Two reference rules are fixed from the development window before that score is computed. The first calls every hold-out day an up day, because up was the majority class in development. The second repeats yesterday's direction. I do not round features to three decimals before fitting. The example did that for a spreadsheet check. The model uses the unrounded values.

Results

What the subset search kept

An intercept-only logistic regression on the development target has an Akaike score of 6,250.3. The best price-level subset scores 6,248.2. The best stationary subset scores 6,246.3. Across 32,767 subsets, the tenth-best price model is only 1.6 points worse than the best, and the tenth-best stationary model is 1.5 points worse. The likelihood barely moves when features are added. The correlations of the selected features with the next day's log return, on the development window, all lie between -0.05 and 0.01.

The price features that clear the frequency rule are yesterday's close, the close three days back, yesterday's high-minus-low, and volume two days back. The stationary features are yesterday's range relative to the close, and the short moving average of past returns. Figure 2 shows the price-level counts. Close lags and yesterday's range dominate the best subsets. The moving averages of the price level almost never appear, which is consistent with those series being smooth copies of the same non-stationary close.

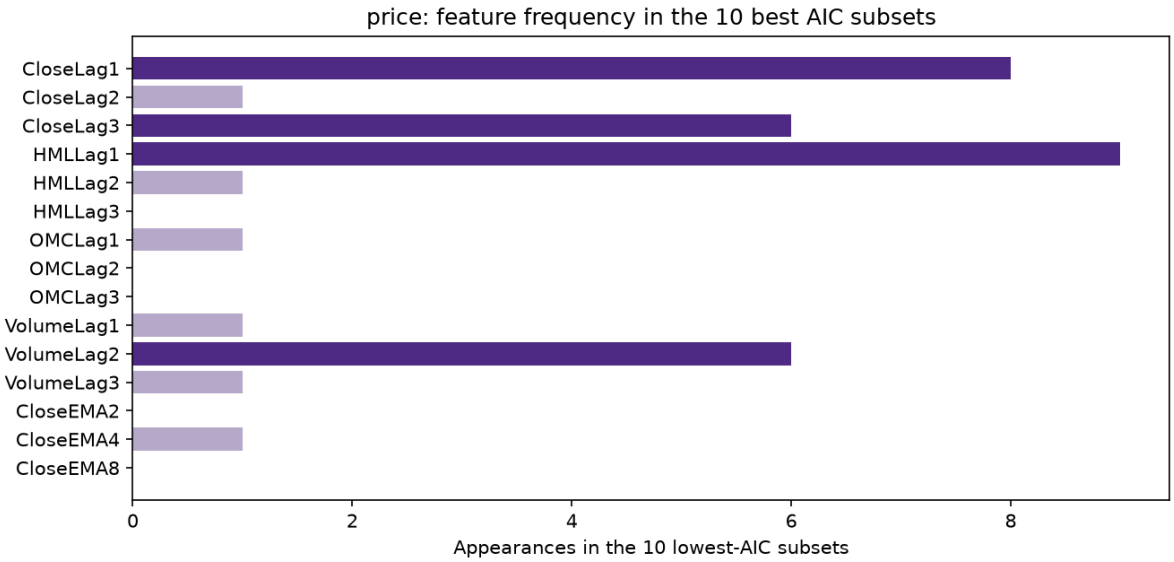


Figure 2. How often each price-level feature appears in the ten lowest-AIC subsets. Dark bars are the four features kept for the classifier.

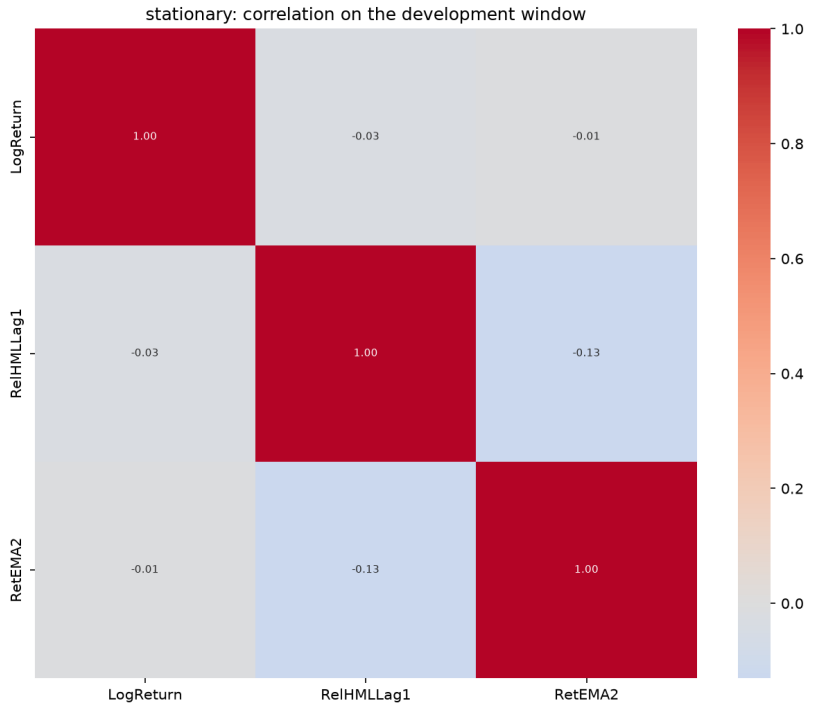


Figure 3. Development-window correlations for the two stationary features and the next day's log return. Both features are nearly uncorrelated with the target.

Cross-validation already said the fit was thin

Model	CV accuracy	CV AUC	In-sample accuracy	In-sample AUC
Price levels	0.505	0.510	0.577	0.616
Stationary	0.509	0.513	0.604	0.662

Table 1. Development-window scores. CV accuracy is the untuned 200-tree model. CV AUC is the tuned model. In-sample scores refit that tuned model on all development rows.

The tuned price model uses depth 2, 138 trees, a learning rate of 0.062, a child-weight minimum of 6, and a subsample of 0.80. The tuned stationary model uses depth 4, 296 trees, a learning rate of 0.013, a child-weight minimum of 4, and a subsample of 0.84. Cross-validated AUC is 0.510 and 0.513. That is a coin flip, computed before the hold-out is opened. The same models, scored on the rows they were fit to, reach AUC 0.616 and 0.662 and accuracy 57.7 percent and 60.4 percent. Those in-sample numbers are the kind of result the jump-start reports for WTI. They describe the training rows.

The hold-out

Rule	Accuracy	AUC
Always up (development majority)	0.569	n.a.
Repeat yesterday's direction	0.458	0.448
Price-level XGBoost	0.432	0.532
Stationary XGBoost	0.552	0.501

Table 2. Hold-out scores. Each model was fit on the development window and applied once. Accuracy uses a probability cutoff of one half.

The price-level model calls a down day on 693 of the 696 hold-out sessions. Its mean predicted probability of an up day is 0.52 on the development rows and 0.36 on the hold-out. The trees split on price levels from a window whose highest close was $193.89. Ninety-four percent of hold-out closes are above that number, and the model reads those new levels as down days. The ROC is only slightly above the diagonal (AUC 0.532), so the ranking contains little information, and the cutoff of one half turns that weak ranking into a one-sided call.

The stationary model does not collapse in that way. Its mean up-day probability is 0.52 in development and 0.53 in the hold-out. Accuracy is 55.2 percent, which sits next to the 56.9 percent from calling every day an up day, and the AUC is 0.501. Repeating yesterday's direction is right on 45.8 percent of hold-out days. Neither fitted model beats the majority rule.

	Price down	Price up	Stationary down	Stationary up
Actual down	299	1	74	226
Actual up	394	2	86	310

Table 3. Hold-out confusion counts at a probability cutoff of one half. The price-level model almost never calls an up day.

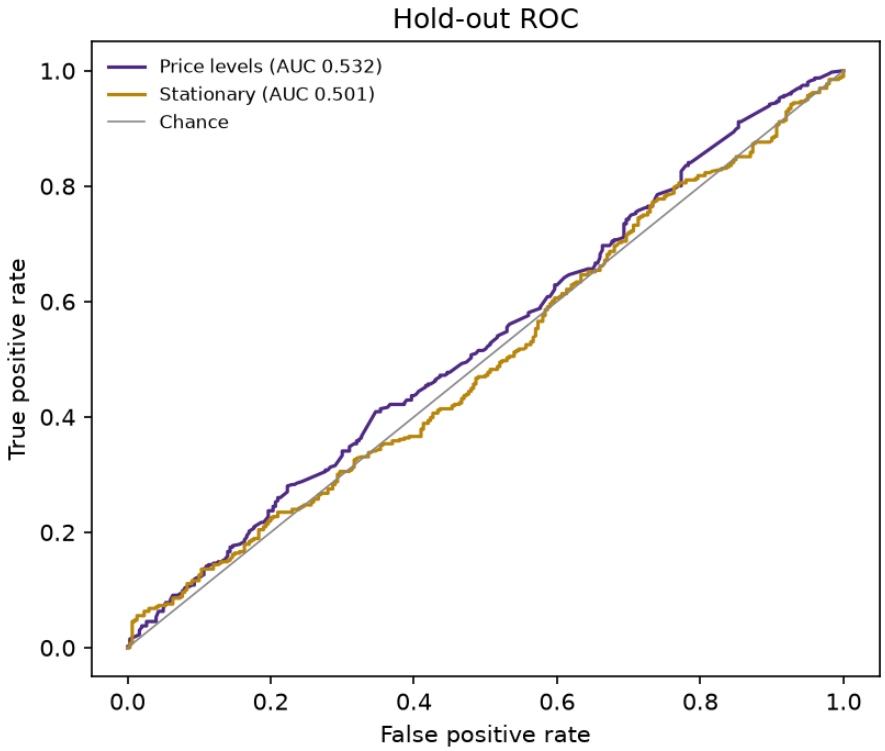


Figure 4. Hold-out receiver operating characteristic curves. Both sit close to the diagonal.

Conclusion

For a trading desk this is a stop, not a position. Over 2024-2026, a next-day long or short rule built from these lagged prices loses to the simpler rule of treating every day as an up day, before commissions, spreads, or the cost of being wrong on a large move. The in-sample accuracy near 60 percent is what a design without a hold-out would have published. The cross-validated AUC near 0.51 already contradicted it. The later window confirmed the cross-validation.

The two families fail differently, and that difference is the stationarity point. The price-level model meets a market that has left its training range, and it calls nearly every new day a down day. The stationary model is calibrated more sensibly and still has no ranking power. A further lag of the same close is unlikely to repair either failure. The module's own next step, other leading series and media measures, is the place to look, and any such series would need the same embargo and the same untouched hold-out.

References

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Note on the use of a language model

A language model in Cursor (Grok 4.7) wrote analysis.py, ran the subset search and the classifier, and drafted this report from output/metrics.json. The session started from the prompt "Assigment module 3." The hold-out window was not used to pick the asset's features or the reported hyperparameters. The numbers in the tables are the program's output.